

Confounders and controls

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Experimental units and independence

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Hypotheses and measurable outcomes

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Methods results and limitations

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Confounders and controls. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Confounders and controls. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Every method A run in morning and B run afternoon; network speed varies by time. Propose a repair.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

New method uses better corpus and different queries while old method uses the original set. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Controls address specified alternatives, not every possible unknown confounder. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Every method A run is morning and B run afternoon; network speed varies by time. Propose a repair.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled confounders and controls evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of confounders and controls. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Confounders and controls

Separate background from class. Training data show every healthy leaf on white paper and every spotted leaf on blue paper.

Goal is leaf condition, not paper colour. A proposed check uses both paper colours for each condition.

Cross class with paper background

	White paper	Blue paper
Healthy	Collect	Collect
Spotted	Collect	Collect

Read a cell using both its row and column labels.

Confounders and controls

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify coupled variables

Condition and background change together in the original collection.



2. Cross the conditions

Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.



3. Inspect the shortcut

Compare predictions when the same leaf's background changes under a controlled test.

CHANGE ONE THING

New cases still use only healthy-white and spotted-blue.

Explain the changed result using the matching lesson.

Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate background from class. Training data show every healthy leaf on white paper and every spotted leaf on blue paper.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Compare predictions when the same leaf's background changes under a controlled test.

1. Identify coupled variables
2. Inspect the shortcut
3. Cross the conditions

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Inspect the shortcut Compare predictions when the same leaf's background changes under a controlled test. A confounder can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

Practice 2: explain the deciding evidence

Changed-case practice: New cases still use only healthy-white and spotted-blue. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. The collection does not resolve the background confounding.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The collection does not resolve the background confounding. The same two variables remain coupled instead of being varied separately.

Repair a missing building block

Review Experimental units and independence (ai-7/9). Goal: Explain the mechanism and terminology of experimental units and independence. Short practice: Count independent plants and photo views. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the mechanism and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

Review Methods results and limitations (ai-7/4). Goal: Explain the mechanism and terminology of methods results and limitations. Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor evidence in their proper categories. Why cannot the result establish that B is best for outdoor farming?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

